EASY · ADVANCED MATH

Nonlinear equations in one variable and systems of equations in two variables

01

10 questions · ~15 min

SAT QUESTION BANK 2026-05-29

Q1 ADVANCED MATH

$$x = 3$$

$$y = 15 - x^2$$

A solution to the given system of equations is x, y. What is the value of xy?

A. 432

B. 54

C. 45

D. 18

Q2 GRID-IN ADVANCED MATH

$$5x = 45$$

What is the positive solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

$$x = 49$$

$$y = \sqrt{x} + 9$$

The graphs of the given equations intersect at the point x, y in the xy-plane. What is the value of y?

A. 16

B. 40

C. 81

D. 130

Q4

ADVANCED MATH

$$3x^2 - 15x + 18 = 0$$

How many distinct real solutions are there to the given equation?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Q5

ADVANCED MATH

$$6x - 9y > 12$$

Which of the following inequalities is equivalent to the inequality above?

A. $x - y > 2$

B. $2x - 3y > 4$

C. $3x - 2y > 4$

D. $3y - 2x > 2$

Q6 ADVANCED MATH

$$p + 61 = 65$$

Which value is a solution to the given equation?

A. $\frac{65}{61}$

B. 4

C. 126

D. 130

Q7

ADVANCED MATH

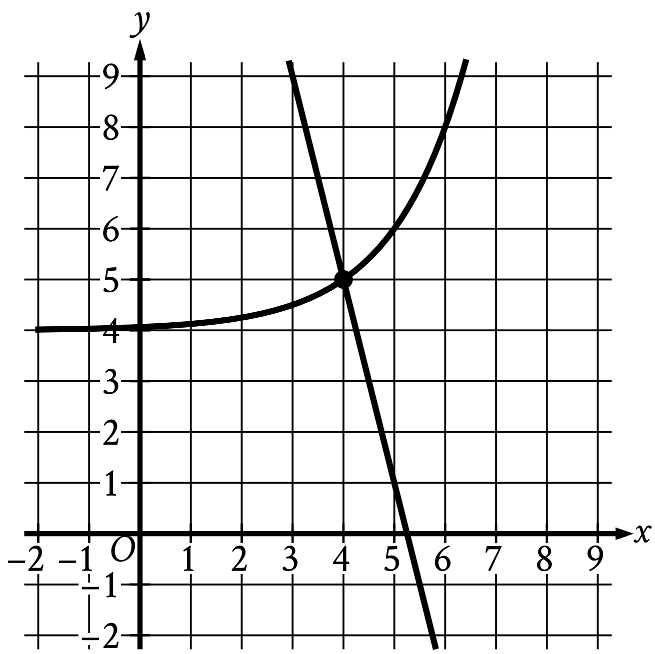
Which of the following is a solution to the equation $2x^2 - 4 = x^2$?

A. 1

B. 2

C. 3

D. 4



- For the curve in the system:
    - The curve passes from quadrant 2 to quadrant 1.
    - In quadrant 2, the curve trends up gradually to the approximate point (0 comma 4.063).
    - In quadrant 1:
        - The curve trends up gradually to point (4 comma 5).
        - The curve then trends up sharply.
    - The curve passes through the following points:
        - (3 comma nine halves)
        - (4 comma 5)
        - (5 comma 6)
- For the line in the system:
    - The line slants sharply down from left to right.
    - The line passes through the following points:
        - (3 comma 9)
        - (4 comma 5)

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

A. $0, 0$

B. $0, 4$

C. $4, 5$

D. $5, 0$

Q9 ADVANCED MATH

If $(x+5)^2=4$, which of the following is a possible value of x ?

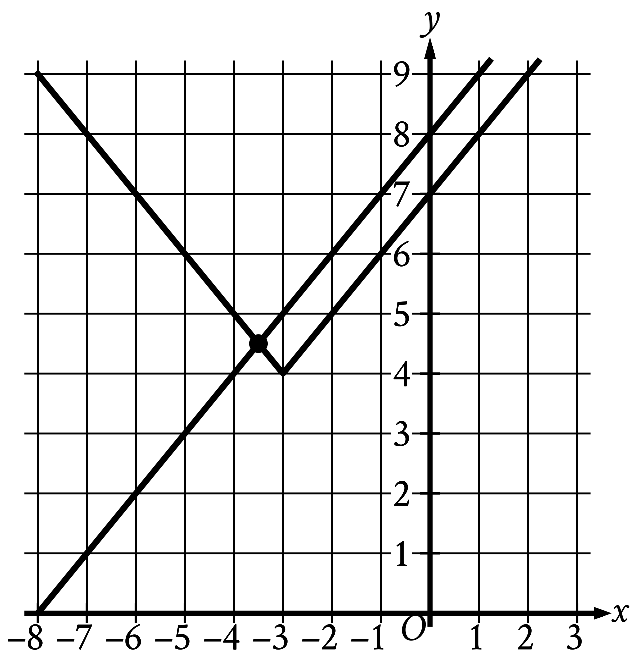
A. 1

B. -1

C. -2

D. -3

Q10 ADVANCED MATH



- For the absolute value function in the system:
  - Moving from left to right:
    - The function slants sharply down to the point (negative 3 comma 4).
    - The function then slants sharply up.
  - The function passes through the following points:
    - (negative seven halves comma nine halves)
    - (negative 3 comma 4)
    - (negative five halves comma nine halves)
- For the linear function in the system:
  - The function slants sharply up from left to right.
  - The function passes through the following points:
    - (negative seven halves comma nine halves)
    - (0 comma 8)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. $0, 8$

B. $\frac{7}{2}, \frac{9}{2}$

C. $-\frac{7}{2}, \frac{9}{2}$

D. $-3, 4$